

- 4 A tube closed at one end, has a constant area of cross section A . Some lead shots are placed in the tube so that the tube floats vertically in a liquid of density ρ . The total mass of the tube and its contents is M .

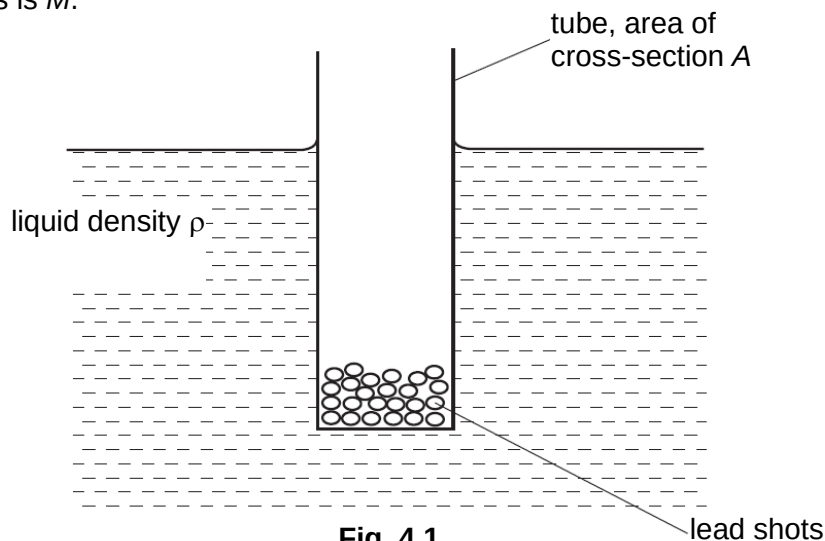


Fig. 4.1

- (a) When the tube is given a small vertical displacement and then released, show that the acceleration a of the tube is related to its vertical displacement y by the expression

$$a = -\frac{A\rho g}{M}y.$$

When in equilibrium and submerge to a depth d , Weight = Upthrust

$$Mg = \rho A d g \quad [\text{M1}]$$

When displaced another depth y , (assign downward positive following downward displacement)

$$Mg - U = Ma$$

$$Mg - \rho A g (d + y) = Ma$$

$$\rho A g d - \rho A g (d + y) = Ma \quad [\text{B1}]$$

$$a = -\frac{A\rho g}{M}y \quad [\text{A0}]$$

[2]

- (b) Fig. 4.2 shows the variation with time t of the vertical displacement y of the tube in another liquid.

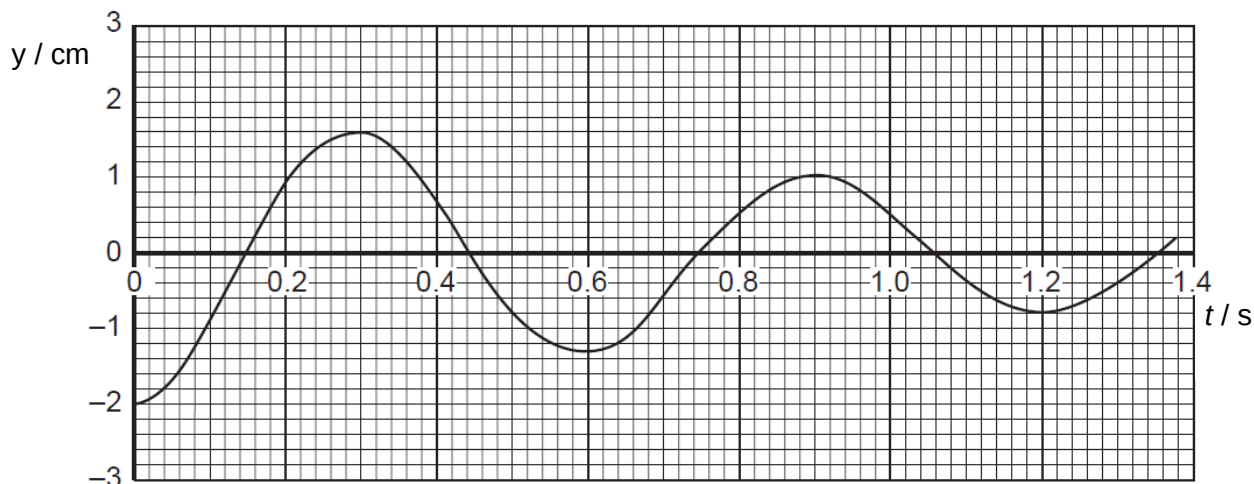


Fig. 4.2

- (i) Determine the frequency f_0 of the oscillating tube.

The tube goes through two oscillations in a time of 1.20 s. [B1]

frequency, $f_0 = 2/1.20 = 1.67 \text{ Hz}$ [A1]

$f_0 = \dots\dots\dots \text{Hz}$ [2]

- (ii) The tube has an external diameter of 2.4 cm and is floating in a liquid of density 950 kg m^{-3} . Calculate the mass of the tube and its contents.

Compare $a = -\frac{A\rho g}{M}y$ with $a = -\omega^2 y$

$$\omega^2 = \frac{A\rho g}{M}$$

$$(2\pi f)^2 = \frac{A\rho g}{M}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{A\rho g}{M}} \text{ [C1]}$$

$$1.67 = \frac{1}{2\pi} \sqrt{\frac{\pi D^2 \rho g}{4 \times M}} = \frac{1}{2\pi} \sqrt{\frac{\pi \times (0.025)^2 \times 950 \times 9.81}{4 \times M}} \text{ [M1]}$$

$$M = 0.038 \text{ kg} \text{ [A1]}$$

mass = $\dots\dots\dots \text{kg}$ [3]

- (iii) More lead shots are added to the tube. State and explain the changes to the graph in Fig. 4.2.

The frequency is reduced with addition of lead shots.

Therefore the graph now shows greater period. [B1: both underlined points]

$\dots\dots\dots$ [1]

[Total: 8]

- 5 Two dippers S_1 and S_2 oscillating in phase with equal amplitude at a frequency of 8.0 Hz